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Digital self-control interventions for distracting media multitasking - A systematic review

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Abstract

Digital distractions can interfere with goal attainment and lead to undesirable habits that are hard to get rid of. Various digital self-control interventions promise support to alleviate the negative impact of digital distractions. These interventions use different approaches, such as the blocking of apps and websites, goal setting, or visualizations of device usage statistics. While many apps and browser extensions make use of these features, little is known about their effectiveness. This systematic review synthesizes the current research to provide insights into the effectiveness of the different kinds of interventions. From a search of the 'ACM', 'Springer Link', 'Web of Science', 'IEEE Xplore' and 'Pubmed' databases, we identified 28 digital self-control interventions. We categorized these interventions according to their features and their outcomes. The interventions showed varying degrees of effectiveness, and especially interventions that relied purely on increasing the participants' awareness were barely effective. For those interventions that sanctioned the use of distractions, the current literature indicates that the sanctions have to be sufficiently difficult to overcome, as they will otherwise be quickly dismissed. The overall confidence in the results is low, with small sample sizes, short study duration, and unclear study contexts. From these insights, we highlight research gaps and close with suggestions for future research.

KEYWORDS

digital distractions, media multitasking, self-control, self-regulation

1 | INTRODUCTION

In learning scenarios such as blended learning or online courses, where learning is largely self-directed, digital content has a Janus-faced role. On the one hand are the potential benefits, such as the access to knowledge resources and the opportunity to exchange information quickly. On the other hand, there are many ways that learners can be distracted from their learning goals, for instance by watching entertaining videos, browsing social networks, or playing video games. Both the goal-congruent and the goal-incongruent

content are often equally accessible. However, while learning can be unpleasant and exhausting, the distracting entertainment is engineered to be fun and addictive, to tempt the users to spend as much time on a platform as possible (Eyal, 2014).

Awareness of the detrimental effects of digital distractions is widespread. When students were asked about their smartphone usage, more than 60% responded that they felt they were overusing their smartphone and that it was a distraction in class (Ko et al., 2015). In an observation of learner behaviour, Rosen et al. (2013) observed that participants switched tasks on average every 6 min, mostly to

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social media. They also observed negative associations between the frequency of switching to social media and the GPA. This negative association between the frequency of using digital distractions during learning and academic performance was observed in several other studies (Kirschner & Karpinski, 2010; Masood et al., 2020).

Different disciplines use different terminology to describe the phenomenon of overuse of distracting content: The term 'media multitasking' is often used in the psychological research literature, and there are validated surveys to measure it (e.g., Baumgartner et al., 2017). While studies find overall that media multitasking is likely to be detrimental to academic performance (van der Schuur et al., 2015), there is still debate about positive effects of media multitasking. Another term that is often used to describe distractions at work is cyberloafing (Varol & Yildirim, 2019). The term cyberloafing describes behaviour that is undesirable, especially from the perspective of third parties such as employers. Whether the behaviour referred to as cyberloafing is bad for the people themselves is, of course, another question. Meanwhile, in the human-computer interaction (HCI) literature, a broad range of terms like 'smartphone non-use' (Hiniker et al., 2016), 'self-interruption' (Kim, Cho, et al., 2017), or 'self-control failure' (Lyngs et al., 2020) is used. Self-control has a clear construct and a model that further sheds light on the mechanisms behind frequent use of distracting content. Self-control intervention is also the term that is used for the digital interventions that are the subject of this review (Lyngs et al., 2019; Schwartz et al., 2021).

2 | THEORY AND RELATED WORKS

If and how often someone follows a short-term gratifying impulse, instead of working towards their long-term goals, depends on several factors that are captured in the trait of self-control. Since this trait is relatively stable (Coyne & Wright, 2014) and even partially hereditary (Willems et al., 2019), some people are more susceptible to disrupting their learning than others.

Another factor that influences the outcome of self-control conflicts is habitual behaviour (Duckworth et al., 2019), which refers to behaviour that has reached a degree of automaticity such that it is no longer directed by conscious goals. It develops through frequent repetition of the same behaviour in a stable context (Fiorella, 2020; Mazar & Wood, 2018; Wood & R nger, 2016). Future behaviour is then initiated by exposure to the same contextual cues, which can be locations, preceding actions, or mental states (Mazar & Wood, 2018; Wood & R nger, 2016). This is especially challenging given the portable and ubiquitous nature of digital devices. Individuals can develop habitual behaviour in a wide range of contexts (Bayer & LaRose, 2018), and it is not unreasonable to expect that many disruptions of learning are due to the habitual use of digital devices. The often-habitual nature of using digital devices can also explain why especially 'heavy users' severely underestimate the amount of time they spend with digital distractions (H. Lee et al., 2014).

For both the conscious and the not-so-conscious uses of distractions, digital self-control interventions aim to provide support where the individual self-control is no longer sufficient. Lyngs et al. (2019)

have conducted a review of consumer-oriented digital self-control interventions from the various app stores. In this review, the authors created an integrative model of self-control and habitual behaviour that aims to explain how these interventions can influence behaviour in the face of distractions (see Figure 1).

As a result of their work, Lyngs et al. (2019) identified 367 different interventions that they sorted into four broad categories, which reflect different targets where these interventions influence the decision to (not) use a distraction. Their categories are: (1) blocking/removing, (2) self-tracking, (3) goal advancement and (4) reward/punishment.

The interventions with features in the category *blocking or removing* block the access to programs or websites entirely, or they set limits on the duration that a user can access distracting content. In some cases, the user can pause the blocking with additional friction, like a menial task or a password. Other interventions in this category remove particularly distracting features, such as a newsfeed or content recommendations, from popular websites. These interventions can prevent habitual use of digital distractions, giving users the chance to reflect on their behaviour and make a conscious decision instead of following an unconscious impulse.

If the behaviour is driven by conscious decisions, the users can benefit from interventions with features from the *self-tracking* and *goal advancement* category. *Self-tracking* interventions display usage statistics or the amount of time that a user spent on distracting activities. Users are supposed to monitor themselves and to decide when they have overused digital content. *Goal setting* interventions remind the users of a task or a goal that they have set for themselves.

Lastly, the category *reward/punishment* contains interventions with features that provide rewards, points, or achievements that users can gain for abstaining from distractions. The rewards or punishments within these interventions are sometimes in the form of social actions in which users support or sanction each other. These features influence the value that users ascribe to resisting using a digital distraction (Shenhav et al., 2013).

While the review by Lyngs et al. (2019) resulted in an extensive overview of existing features, it is not known how effective they are at reducing the negative impact of digital distractions. In the context of self-directed learning with digital media, one can imagine scenarios for the various interventions in which they work well, or less well. For example, the approach of complete blocking does prevent distractions from being used. However, complete blocking also means that a learner can no longer access the recording of a course if it is hosted on the same video platform as the entertainment. For such scenarios, only interventions that still allow access to a platform are usable at all. These more permissive interventions then raise questions, like: How well do they prevent entertainment from being used instead of learning? Is it sufficient that learners can see their usage statistics in order to reduce distractions at critical moments? Are self-set time limit goals adhered to, or are additional restrictions necessary? With this review, we address these gaps by providing a systematic literature review of peer-reviewed publications that studied changes in the use of digital distractions when participants used digital self-control interventions.



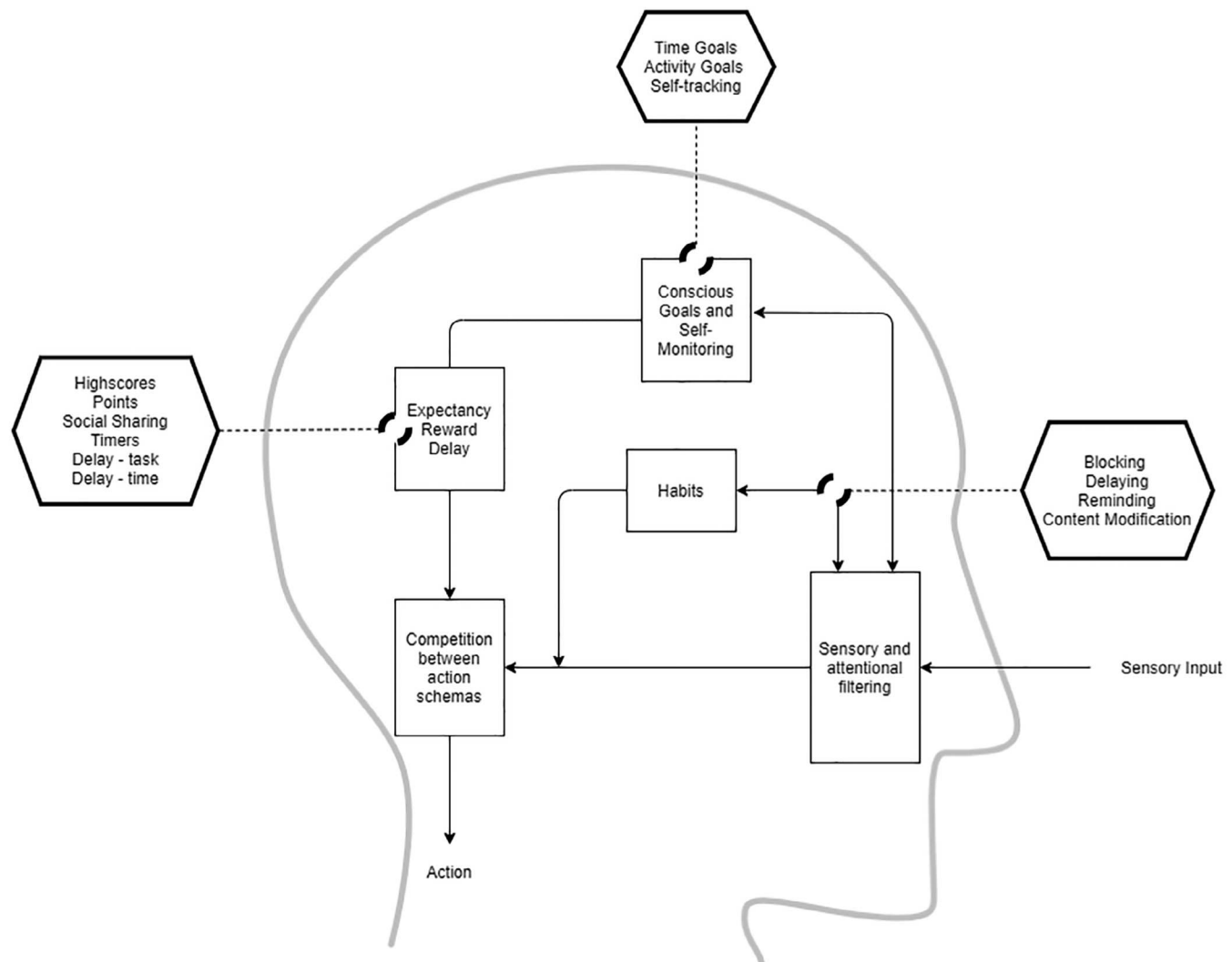


FIGURE 1 An extended dual systems model of self-control that Lyngs et al. (2019) adapted from Shea et al. (2014). The hexagons list interventions that are connected via dashed lines to the targets where they intervene in the process from sensory input to action

3 | THE PRESENT STUDY

In this study, we contribute to the research on digital self-control interventions with a systematic review (Petticrew & Roberts, 2006) where we examine the effectiveness of these interventions and their underlying features. We used the following research questions to guide our study:

RQ 1: Which types of intervention have evidence for achieving changes in the use of digital distractions?

RQ 2: How effective were the interventions at alleviating digital distractions?

To answer RQ 1, we first looked for published research regarding digital interventions for digital distractions. We then categorized the identified interventions based on the categories proposed by Lyngs et al. (2019). We present these results in a narrative form (compare Petticrew & Roberts, 2006, p. 165f.) where we describe the features of the various interventions. To answer RQ 2, we analyzed the reported results and, when possible, calculated the effect

sizes of the interventions (See the Data Extraction and Analysis section).

4 | METHOD

For this review, we followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines (PRISMA) (Moher et al., 2009), which sets out the minimum standards of reporting that should be present in a systematic review, and a flow of information to apply for the study selection.

4.1 | Eligibility criteria

We included publications that reported the effects of digital interventions to alleviate digital distractions. Our inclusion criteria were publications that...

- ...were written in English.
- ...were published in a peer-reviewed outlet.
- ...applied a digital intervention to reduce the use of distractions on a computer, smartphone, or a wearable device.
- ...reported outcome measures related to changes in the use of digital distraction. Since an effect for a reduction in use is guaranteed for interventions that completely block the distractions, we included use of the intervention as an outcome. The rationale is that if the blocking is used more often, distractions can be accessed less frequently.
- ...had a study design that compared outcomes under treatment with the outcomes of a non-treatment group, or a single-group design with a baseline measurement.
- ...investigated a nonclinical population, for example, no special needs or ADHD population.

4.1.1 | Search strategy

We performed a search of the relevant databases and combined these results with a snowballing approach to obtain additional literature from the references and the citing literature (see Figure 2). For the database search, the search concepts were (1) digital distractions and (2) interventions. From the terminologies used in the literature, we arrived at the search term ('self-control' OR 'self-regulation' OR 'mind-wandering' OR cyberloafing OR 'media multitasking' OR willpower OR 'digital distractions' OR 'smartphone addiction') for the topic of digital distractions. For the interventions, the terms were (intervention* OR app OR browser* OR tracker OR self-monitoring OR digital OR smartphone). Both terms were combined with a logical AND. We searched the databases with the broadest applicable search scope. The databases were: 'Web of Science' (scope: Topic), 'Springer Link'

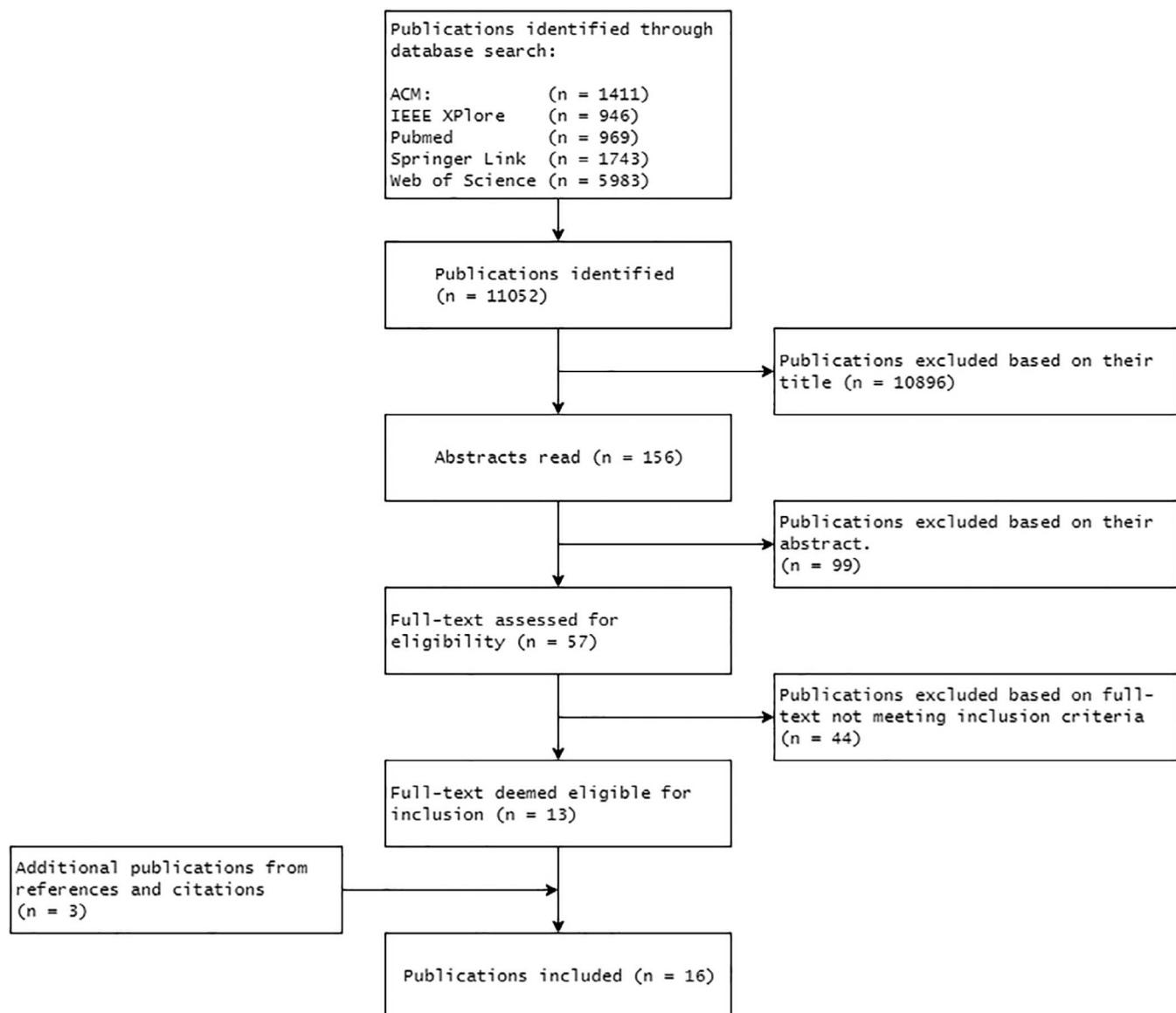


FIGURE 2 Flowchart of the paper inclusion process

(scope: All), 'ACM Digital Library' (scope: Anywhere), 'IEEE Xplore' (scope: Full Text and Metadata) and 'Pubmed' (scope: All Fields). We copied all the results into a spreadsheet, where the publications were filtered first by title. Next, we read the abstracts and, for those publications which appeared eligible based on their abstract, we read the full text. From the results of the database search that we considered as relevant, we looked for further papers by incorporating the referenced literature and the literature which cited the publication. For this step, we used 'Google Scholar', which had the most extensive results for citing literature.

4.2 | Data extraction and analysis

For each publication, we extracted a description of the intervention(s), the research questions or hypotheses, the outcomes, the year of publication, the study design, the study population, recruitment criteria, the study duration, the target distraction, and the measured effects of the intervention(s). If a publication contained multiple interventions or multiple studies for one intervention, we extracted the parameters mentioned above separately for each intervention and each study.

To categorize the features, the first and second authors of the paper independently rated the interventions in the publications according to the coding criteria. The inter-rater agreement between authors was 91%. Consensus about the remaining 9% of the categorizations was reached after a brief discussion.

To calculate effect sizes and confidence intervals, we used the following versions of the calculation of standardized mean difference *Cohen's d*. For between subjects-design we calculated the difference of the means (*M*) divided by the average standard deviation (*SD*) (Lakens, 2013):

$$\frac{M_1 - M_2}{\left(\frac{SD_1 + SD_2}{2}\right)}$$

For within-subjects design, we calculated the effect size as the difference of the means divided by the pooled *SD* (Lakens, 2013):

$$\frac{M_1 - M_2}{\sqrt{\frac{(n_1 - 1)SD_1^2 + (n_2 - 1)SD_2^2}{n_1 + n_2 - 2}}}$$

For publications that did not allow conversion to these effect size measures because of missing data in their report, we tried to obtain the data by contacting the corresponding authors of the publication.

4.3 | Risk of bias and quality assessment

To assess the study quality and the risk of bias in the individual studies, we followed the study quality assessment tools by the NHLBI.¹ In addition to the criteria from those checklists, we considered study duration, the conduct of a follow-up, and comprehensible reporting of

results. This assessment only determines how well the studies are suited to answer our research questions, and does not represent a general categorization into 'good' or 'bad' studies.

4.4 | Search results

We found 16 publications eligible for this review, which contained a total of 28 interventions. The interventions are listed with a brief description in Table 1. We assigned an identifier consisting of the number sign # and a number to each intervention (#1–#16). Nine publications contained more than one intervention and we appended a lower-case alphabetic letter to their id (e.g., #7a, #7b).

5 | FINDINGS AND ANALYSIS

We start this section with a high-level overview of the interventions and the outcomes that the studies report, followed in the subsections by a narrative description of the interventions. Regarding the multiple interventions within one publication, it should be noted that some interventions within one publication often differed only in minor details, such as the time spent on a distraction before an intervention started. With a few exceptions, we categorized the interventions following the taxonomy that was identified in the review of the commercially available interventions (Lyngs et al., 2019) that we describe above. We deviated in the two following categories: First, we use the term 'awareness' (cf. Parry & le Roux, 2019) instead of 'self-tracking' because the concept of increasing the users' awareness is more appropriate. Second, we placed interventions that modified the features or content into their own category instead of together with the blocking features. Blocking is functionally different from the modification of features because blocking completely prevents access to the distraction. Feature modification comes into effect only after the content has already been accessed, and a change in activities has already taken place.

The interventions monitored and targeted different kinds of digital activities on different devices. The monitored activities were either smartphone apps, computer programs, or websites (see Table 3). The monitoring was mostly limited to the start of the activity and the duration with which it was active. In one publication, the analysis was extended to more fine-grained interactions within the distraction^{#16a, #16b}. The interventions were active on either mobile devices (primarily smartphones) and computers. Only one intervention synchronized the monitoring across smartphones and computers. From the interventions on mobile devices, six of the interventions monitored all app activity, nine monitored activity only for apps from a user- or application-defined list, and two interventions monitored only the activity in one specific app. All of these interventions that ran on a computer were designed to monitor website activity, and four interventions also monitored the programs that the participants used. Eleven of the interventions which monitored the browser activity used specific user- or application-defined URLs as intervention

TABLE 1 The publications that were included in this review and a description of the respective interventions

ID	References	Description
#1	Lottridge et al. (2012)	Browser redesign to frame tabs either as 'work' or 'non-work'. 'Work' tabs were larger, while other tabs were smaller and placed on the right side. A 'work versus non-work' ratio progress bar was added to the browser status bar.
#2a #2b	Ko et al. (2015)	NUGU - A smartphone app that encouraged participants to set time goals during which they commit to not use their smartphone. One group had social comparison features enabled (#2a) and the other group did not (#2b).
#3a #3b	Foulonneau et al. (2016)	TILT - A smartphone app that displayed device use in several visualizations: messages, notifications and home screen widgets. The content of the visualizations was personalized (#3a) or static (#3b).
#4	Hiniker et al. (2016)	MyTime - An app to set time limit goals for monitored apps. Upon reaching those goals, reminders to stop using those apps popped up.
#5a #5b	Y.-H. Kim et al. (2016)	TimeAware - Information display of time spent on the device with either a productivity (#5a) or a distraction (#5b) framing.
#6	Terry et al. (2016)	Daily text messages with facts about the negative effects of media multitasking.
#7a #7b	Whittaker et al. (2016)	meTime - An always-on dashboard that showed the last used applications. Longer use of an application resulted in a larger display of this application in the dashboard.
#8	Kim, Cho, et al. (2017)	PomodoLock - Distracting apps and websites were blocked for 25-min intervals on both smartphone and PC.
#9a #9b	Kovacs et al. (2018)	HabitLab - A browser extension with many different interventions. The study with intervention #9a investigated the effect of rotating interventions on attrition, and study #9b the effect of providing additional information and control over the interventions.
#10a #10b	Okeke et al. (2018)	Repeating smartphone vibrations when a monitored website was visited longer than allowed (5 min #10a, personalized #10b).
#11a #11b #11c	Kim, Park, et al. (2019)	Lock n' Type - Participants had to enter a string of random digits before opening monitored apps. Digit input lengths were zero (#11a), 10 (#11b), and 30 (#11c).
#12a #12b #12c	Kim, Jung, et al. (2019)	Goalkeeper - different lockout mechanisms based on individual time goals. From reminder notifications (#12a), to successively longer lockout durations (#12b), and full-day phone lock upon exceeding the time (#12c).
#13a #13b	Tseng et al. (2019)	UpTime - Chatbot-based website blocker (#13a) compared with a time-based website blocker (#13b). With #13a, users had to provide reasons for visiting distracting websites.
#14	Holte and Ferraro (2020)	Changing the display colours of Smartphones to grayscale to decrease the attractiveness of distracting content.
#15	Loid et al. (2020)	Evaluation of the effect of push notifications on screen time and phone checking behaviour.
#16a #16b	Lyngs et al. (2020)	Two alterations of the Facebook website. In #16a, users had to set action goals when entering the site; In #16b, the newsfeed was removed.

targets, while two interventions monitored only the activity on a specific website. The following subsections give more details regarding the description of the intervention features and the outcomes that were achieved (Figure 3).

5.1 | Description of intervention features

5.1.1 | Awareness features

Interventions with awareness features inform the participants about their use of distracting activities. This information is presented with the aim to start a process of self-reflection so that the participants themselves decide to spend less time on digital distractions.

Awareness interventions varied according to (a) the mode in which they presented the information to the participants, whether it

was within a program or an app, or via instant message, or as a push notification. (b) the content that was presented to the participants, whether it was statistics, messages, or graphics.

The most common form of presentation was the display of usage statistics: Whittaker et al. (2016) created a computer program^{#7a,#7b} which displayed the most-used programs from the last 30 min to the participants, showing the most-frequently used programs larger than the less frequently used ones. Lottridge et al. (2012) modified the web browser to show statistics about the frequency of work versus non-work websites that the participants visited^{#1}. Other interventions contained visualizations of usage statistics on top of various other features^{#2a,#2b,#12a,#12b,#12c} that are described in the following sections.

Another common form of presentation were different kinds of notifications that alerted the participants: text messages^{#6}, push notifications^{#3a,#3b,#15a}, and popup message dialogues^{#11a,#4,#12a}.



ID	Device	Monitored Activities	Feature Categories					Outcome Categories			
			Aware	Block	Goals	Rewards & Social	Content Mod.	Time on distraction	Distraction start	Time on device	Intervention use
1	☑	Specific websites						+	+		
2a	☐	All apps						+	+	+	+
2b	☐	All apps							+	n.s.	n.s.
3a	☐	All apps								n.s.	
3b	☐	All apps								n.s.	
4	☐	Specific apps						n.s.			
5a	☑	All websites, programs						n.s.			n.s.
5b	☑	All websites, programs						+			n.s.
6	☐	Specific apps						+			
7a	☑	All websites, programs						+		+	
7b	☑	All websites, programs						+		+	
8	☑ / ☐	Specific websites, apps									n.s.
9a	☑	Specific websites						+			-
9b	☑	Specific websites									+
10a	☐	Facebook app						+			
10b	☐	Facebook app						+			
11a	☐	Specific apps						-	+	n.s.	
11b	☐	Specific apps						-	+	n.s.	
11c	☐	Specific apps						-	+	n.s.	
12a	☐	Specific apps								n.s.	
12b	☐	Specific apps								+	
12c	☐	Specific apps								+	
13a	☑	Specific websites						+	+		+
13b	☑	Specific websites						n.s.	n.s.		-
14	☐	All apps, websites						+		+	
15	☐	All apps, websites							n.s.	n.s.	
16a	☑	Facebook website						+	+		
16b	☑	Facebook website						n.s.	+		

FIGURE 3 Parameters that describe the various interventions. The ‘Device’ column lists the device type on which the intervention was active (computer or mobile device). The columns on monitored activities and feature categories list these parameters for each intervention. The columns under the ‘Outcome Categories’ header list the outcomes that were achieved with the intervention. A ‘+’ signifies that the intervention had significant positive results on this outcome, a ‘n.s.’ that there were no significant effects on this outcome, and a ‘-’ stands for negative effects on this outcome

The most insistent awareness intervention, created by Okeke et al. (2018), started smartphone vibrations when a time quota for a distracting website (in their case, Facebook) was exceeded^{#10a,#10b}. The smartphone vibrated until the user left the distracting website.

With two exceptions, the interventions informed the participants about the extent of distracting activity use. The first exception were the SMS notifications in the intervention from Terry et al. (2016), which contained facts about the negative effects of media multitasking^{#6}. The second deviation concerned the framing of the content, where all but one intervention focused on the distracting activities. To investigate the effect of a positive framing, Y.-H. Kim et al. (2016) used visualizations for the use of productive programs like word processors^{#5a}, and compared this with distraction-focused visualizations^{#5b}.

5.1.2 | Goal-advancement features

We considered a feature as goal-advancement if participants had the option to set goals related to a reduction of time spent on digital distractions. The goals were either time-based or action-based, where the participants could enter their goals in a free text. The majority of the interventions with goal-advancement features

asked the participant to set a time limit for their distracting activities^{#2a,#2b,#4,#12a,#12b,#12c}, and free text goals were less common^{#16a,#4}. To sanction goal deviations, the interventions used blocking of distractions^{#12b,#12c}, repeated reminders^{#16a,#12a,#4}, or the reduction of a virtual score^{#2a,#2b}.

Kim, Jung, et al. (2019) investigated sanctions of exceeding time-goals with three different interaction lockout mechanisms (ILM): In the *Non-ILM* condition^{#12a}, users received only notifications that reminded them of their time limit. In the *Weak-ILM* condition^{#12b}, the participants' phones were locked for increasing durations, and *Strong-ILM*^{#12c} locked the users' phones for the rest of the day after exceeding their time limit goal.

Hiniker et al. (2016) evaluated a smartphone app which supported goals that were both time- and action-based^{#4}. Participants were asked to enter a time limit goal and an optional free text goal for the use of monitored apps. After exceeding the self-set time limit, the participants were reminded of the excess with a warning prompt. Following this prompt, they could close the app, request a time extension, or simply dismiss it.

Lyngs et al. (2020) modified the *Facebook* homepage such that participants were prompted to enter free text action goals on each visit^{#16a}. The goal that the participants entered was periodically repeated during their visit on the website.

5.1.3 | Blocking features

An intervention was considered as containing blocking features if it prevented access to the distraction. For blocking interventions, we found interventions where the blocking was rigorous^{#8,#10,#12b,#12c,#13b} and others where lifting of the block could be negotiated^{#11b,#11c,#13a,#12b}.

Mark et al. (2018) blocked distractions during work time with the 'freedom' software, a blacklist-based website blocker^{#10}. Other interventions were less rigorous and blocked distractions only for specific time intervals. Kim, Cho et al. (2017) created a *Pomodoro*-timer intervention on the usage of distracting websites. In the *Pomodoro* method, users set a time interval (typically 25 min), during which they plan to work productively. After this time interval, they get a break, typically 5 min, during which they are free to do whatever they like, before starting the next *Pomodoro* session.

In two publications, the block was not absolute, and could be lifted: Tseng et al. (2019) created the *UpTime*-system^{#13a}, a website-blocking system with a chatbot to control a time-based blocking system for the transition from break to work. A break was operationalized as an absence of user input for at least 5 min. Upon resumption of work, the chatbot was activated and started disabling access to distracting websites for the next 25 min. The participants could negotiate a lift of the block by navigating through a conversation with the chatbot. The chatbot also suggested to start blocking sessions when a participant spent more than 15 min on distracting websites. This system was compared with a time-based blocking system^{#13b} (cf. #8). Kim, Park, et al. (2019) created the *Lock n' Type* intervention, which required participants to enter a string of random digits on their smartphone before they could open a blacklisted apps. They tested three different variants of this intervention, which each required different lengths of random digits^{#11a,#11b,#11c}.

5.1.4 | Modification of content

For this review, features were considered as 'Modification of Features or Content', if they modified certain aspects of a digital distraction to make it still usable, but less appealing to the participants. To achieve this, these interventions remove non-essential parts of the distraction that are typically designed to convince the users to spend more time with an app or a website.

Holte and Ferraro (2020) switched their participants' smartphone displays to grayscale^{#14} in order to reduce gratification from distracting activities. The grayscale filter was an integrated feature of their participants' smartphones, and the filter was active at all times and for all apps. Lottridge et al. (2012) modified the web browser tab bar^{#1} so that tabs that were classified as work-related had their colour contrast enhanced, and were made larger. Tabs from non-work URLs were made smaller and always displayed on the right.

Kovacs et al. (2018) report about 'Habitlab', a browser extension that rotates behaviour change interventions for distracting websites, that is, the participants experienced different interventions between visits to the same site^{#9a,#9b}. The authors are not specific about all of

the interventions that were active in Habitlab at the time of their studies, but they do mention a few example interventions, like a news feed blocker for Facebook, or hiding 'related videos' on YouTube.

Lastly, Lyngs et al. (2020) used a browser extension to remove the newsfeed^{#16b} from the Facebook homepage.

5.1.5 | Reward and punishment

Ko et al. (2015) added social support and point reward features to their 'NUGU' smartphone app. The participants were asked to specify a duration for which they wanted to stop using their smartphone (e.g., 'studying for 30 minutes'). These limiting sessions were conceptualized as 'missions', and completing a mission gave the users virtual points. If a participant used the smartphone for anything except incoming calls, the mission failed. Participants could also form groups, start missions together, and compare their time-limiting efforts with those of their peers^{#2a}.

5.2 | Outcomes

For the outcome measurements, we clustered the outcomes into the categories (a) time that participants spent on the distraction, (b) the frequency with which a participants started a distraction, (c) the total time that participants spent on the device, and (d) measures of using and interacting with the intervention.

Measures for the time that participants spend with the distraction are a straightforward way to gauge the effectiveness of an intervention. In this category, we included outcomes that measured how much time the participants spent on websites or in apps that were marked as a distraction by the participants themselves or by the researchers. Summarized in Table 2, the time spent on distraction was measured for 17 interventions.

Measures of the frequency with which distractions are started subsume measures such as navigating to a URL, starting an app, or unlocking the phone. Changes in the frequency of distraction starts were measured for nine interventions, summarized in Table 3.

Time spent on device measures does not discriminate between different activities on the device and can also include the use of apps or programs related to work or learning. These measures include the time spent across all apps or on all URLs. Time spent on the device was measured for 16 interventions. The results are summarized in Table 4.

Measures of how a participant interacts with an intervention can also provide indications of the effectiveness of an intervention. For blocking interventions, an active intervention means that no distracting activity can be started in the first place. Thus, a participant that activated such an intervention more often, might have made the intervention more effective than a participant that rarely activated the intervention. These measures can, furthermore, hint as to whether the intervention failed because it was generally not suited to achieve the task, or because the participants simply did not use it. The results for this outcome category are summarized in Table 5.



TABLE 2 The measures that we subsumed under the category time that participants spent on the distraction

ID	Outcome	Measure	Effect size [95% CI]
#1	Time on distraction	Time in non-work URLs	(+)
#4	Time on distraction	Time in monitored apps	(+)
#5a	Time on distraction	Time in monitored apps/URLs	n.s.
#5b	Time on distraction	Time in monitored apps/URLs	(+)
#6	Time on distraction	MTUAS Scale	n.s.
#7a	Time on distraction	Time on Facebook	0.85 [0.47, 1.22]
#7a	Time on distraction	Browsing time	0.74 [1.11, 0.37]
#7b	Time on distraction	Time on Facebook	(+)
#9a	Time on distraction	Time in monitored URLs	(+)
#10a	Time on distraction	Time on Facebook	0.49 [−0.23, 1.22]
#10b	Time on distraction	Time on Facebook	0.60 [−0.05, 1.25]
#11a	Time on distraction	Time in monitored apps	(−)
#11b	Time on distraction	Time in monitored apps	(−)
#11c	Time on distraction	Time in monitored apps	(−)
#13a	Time on distraction	Time in monitored URLs	0.14 [−1.04, 1.33]
#13b	Time on distraction	Time in monitored URLs	n.s.
#14	Time on distraction	Use of social media apps	0.44 [0.11, 0.77]
#14	Time on distraction	Use of video player	n.s.
#14	Time on distraction	Use of web browser	0.46 [0.13, 0.78]
#15	Time on distraction	Self-report	n.s.
#16a	Time on distraction	Time on Facebook	1.62 [0.90, 2.33]
#16a	Time on distraction	Facebook visit length	n.s.
#16b	Time on distraction	Time on Facebook	n.s.
#16b	Time on distraction	Facebook visit length	0.75 [0.09, 1.40]

Note: When the reporting did not allow the conversion into Cohen's *d*, the measure is marked with a (+) if a significant reduction in the time that participants spent on the distracting activity was reported. No significant effects for this measure are marked with a 'n.s.', and instances where an increase in the time spent on the distracting activity was observed are marked with a (−).

TABLE 3 The measures that we subsumed under the category frequency of distraction starts

ID	Outcome	Measure	Effect size [95% CI]
#1	Distraction start	Navigation to monitored URLs	1.84 [0.56, 2.22]
#2a	Distraction start	Opening apps	0.56 [0.09, 1.04]
#2b	Distraction start	Opening apps	n.s.
#11a	Distraction start	Opening monitored apps	(+)
#11a	Distraction start	Discouraged app starts	(+)
#11b	Distraction start	Opening monitored apps	(+)
#11b	Distraction start	Discouraged app starts	(+)
#11c	Distraction start	Opening monitored apps	(+)
#11c	Distraction start	Discouraged app starts	(+)
#13a	Distraction start	Periods with visits to monitored URLs	0.64 [−0.56, 1.84]
#13b	Distraction start	Periods with visits to monitored URLs	n.s.
#15	Distraction start	Number of phone checks	n.s.
#16a	Distraction start	Number of visits to Facebook	1.62 [0.90, 2.33]
#16b	Distraction start	Number of visits on Facebook	n.s.

Note: When the reporting did not allow the conversion into Cohen's *d* but still reported significant reductions, the entry is marked with a (+). No significant effects are marked with a 'n.s.'.



5.2.1 | Awareness intervention effects

Awareness interventions led to reductions in time spent on distractions and time on the device both through the display of usage statistics^{#5a,#5b,#7a,#7b} and notifications^{#10a,#10b}.

The interventions #7a and #7b achieved a medium to large effect for time on distraction and total time on the device, but both interventions were active for only 2 days (see also Table 6). Dashboard interventions led to less time on distraction only with distraction-framed content^{#5b}, but not with a productive framing^{#5a}. Neither distraction nor productivity framing^{#5a,#5b} made a difference regarding how long and how often the participants accessed the information.

Awareness notifications only achieved positive effects when they were insistent. For usage monitoring via notifications, no effect was observed on total time on device^{#3a,#15}, and frequency of distraction starts^{#15}. Regular messages about the negative effects of excessive media consumption also had no positive effects^{#6}. Only the smartphone vibrations to remind of overuse led to less time spent on the distraction with personalized^{#10a} ($d = 0.49$) and static time quotas^{#10b} ($d = 0.69$).

5.2.2 | Goal-advancement intervention effects

Time-based goal setting with warning prompts led to reductions in time on device, and the amount of time spent on distractions in one intervention^{#4}, while in other interventions, these prompts of exceeding a time limit did not result in a reduction in time on device until the excess was sanctioned^{#12a}. When sanctions in the form of device lockout^{#12b,#12c} were added, time on device was reduced (#12b: $d = 0.35$, #12c: $d = 0.54$).

When participants set action goals in addition to time goals, they were more likely to follow prompts to leave a distraction^{#4}. Setting action goals immediately before the start of a distracting activity^{#16a} also led to a large reduction in total time spent on the distraction ($d = 1.62$) and the frequency of distraction starts ($d = 1.62$). Changes in the duration of individual sessions were not significant.

5.2.3 | Blocking intervention effects

Blocking the access to distractions with a task convinced participants to not start distractions proportionally to the task difficulty: Requiring a 30-digit input before opening an app^{#11c} was more effective ($d = 0.47$) than a 10-digit input task ($d = 0.27$) or a simple warning prompt ($d = 0.13$). This task-based delay was the only type of intervention for which an increase in time on distraction was reported, indicating that participants compensated for the initial hurdle of opening an app by spending more time in it. Still, the aspect of being able to negotiate the lift of a block was preferred^{#13a} to time-based blocking, and participants activated the intervention more often ($d = 2.06$) and started distraction less often ($d = 0.82$) than with purely time-based blocking intervention^{#13b}. However, the chatbot also suggested the initiation of additional blocking sessions, thereby potentially skewing the results.

A measure that is particularly relevant for blocking interventions is how frequent the blocking is activated. Participants in a blocking condition^{#8} completed more limiting sessions in the first week than a control group which only monitored themselves. Over the full two-week duration, these differences were no longer observable.

5.2.4 | Content modification intervention effects

Setting the smartphone display to grayscale^{#14} led to a reduction in total phone use ($d = 0.39$), social media use ($d = 0.44$), and internet browser use ($d = 0.46$). No effect on video player use was found.

Removing the Facebook newsfeed^{#16b} resulted in a reduction for the duration of an individual session ($d = 0.75$). No effects were found for the total time spent on the distraction or the starts of the distracting activity.

5.2.5 | Reward intervention effects

When social support features were added to a goal-advancement intervention^{#2a}, participants set significantly more time-limit goals per day ($d = 1.87$), reduced their time on the device ($d = 0.83$), and started distractions less often ($d = 0.56$). These changes were not observed in the variant without the social support features^{#2b}.

5.2.6 | Effects of interventions with mixed feature contribution

For some of the interventions with multiple features, the effect of the intervention cannot be assigned to a single feature. The browser redesign^{#1} achieved reductions for the time on distraction and the start of distractions. However, during the intervention period there were fewer URLs total visited ($d = 1.39$), and fewer total tabs open in general, indicating that there was an overall reduction in activity. Rotating different behaviour change interventions led to a reduction in time spent on distraction^{#9a}. The rotation of features was also the focus of the most thorough investigation of intervention use. Kovacs et al. (2018) found that rotating the active interventions led to less time on distraction, but also to more attrition, that is, uninstalling the intervention^{#9a}. The additional information and the additional options^{#9b} that were granted to participants in the follow-up study reduced the attrition significantly. While in the control condition, only 44% of the participants remained after 7 days, the additional information led to 79% survival, and the additional options condition to 80% survival.

5.3 | Confidence in the results and study quality

In this section, we present our assessment of the study quality and the confidence that we have in the results of the studies. We start with the sampling criteria and the sample size, followed by the study duration,



TABLE 4 The measures that we subsumed under the category time that participants spent on the monitored device

ID	Outcome	Measure	Effect size [95% CI]
#2a	Time on device	Time in all apps	0.83 [0.34, 1.32]
#2b	Time on device	Time in all apps	n.s.
#3a	Time on device	Time in all apps	n.s.
#3b	Time on device	Time in all apps	n.s.
#4	Time on device	Average daily smartphone use	(+)
#7a	Time on device	Total time on PC	0.61 [0.24, 0.97]
#7b	Time on device	Total time on PC	(+)
#11a	Time on device	Total time on smartphone	n.s.
#11b	Time on device	Total time on smartphone	n.s.
#11c	Time on device	Total time on smartphone	n.s.
#12a	Time on device	Smartphone use on workdays	n.s.
#12a	Time on device	Smartphone use on weekends	0.56 [0.13, 0.99]
#12b	Time on device	Smartphone use on workdays	0.35 [−0.12, 0.82]
#12b	Time on device	Smartphone use on weekends	0.51 [0.09, 0.93]
#12c	Time on device	Smartphone use on workdays	0.54 [0.07, 1.01]
#12c	Time on device	Smartphone use on weekends	0.51 [0.10, 0.96]
#14	Time on device	Total time on smartphone	0.39 [0.06, 0.72]
#15	Time on device	Total time on smartphone	n.s.

Note: When the reporting did not allow the conversion into Cohen's *d*, the measure is marked with a (+) if a significant reduction of the time that the participants spent on the device was reported. No significant effects for this measure are marked with a 'n.s.'.

TABLE 5 The measures subsumed under the category use of the intervention

ID	Outcome	Measure	Effect size [95% CI]
#2a	Intervention use	Number of goals	(+)
#2b	Intervention use	Number of goals	n.s.
#5a	Intervention use	Dashboard access	n.s.
#5a	Intervention use	Widget access	n.s.
#5b	Intervention use	Dashboard access	n.s.
#5b	Intervention use	Widget access	n.s.
#8	Intervention use	Number of started sessions	n.s.
#9a	Intervention use	Installation survival	(−)
#9b	Intervention use	Installation survival	(+)
#13a	Intervention use	Number of sessions	2.06 [1.17, 2.94]
#13b	Intervention use	Number of sessions	n.s.

Note: Significantly more interactions with the intervention are marked with a (+). No significant effects for this measure are marked with a 'n.s.'.

For the study sample we found several recruiting criteria that were mentioned in the publications (see column 'Selection Criteria' in Table 6). These included participants with an interest in reducing smartphone usage, participants that felt that they were often distracted, or interested in improving productivity. In the study (Lottridge et al., 2012), participants were pre-screened with the Media Multitasking Index (Ophir et al., 2009), and only those who scored either high or low in this index were included. Kim, Jung, et al. (2019) selected participants according to their readiness within the Transtheoretical Model framework (Prochaska & Velicer, 1997). None of the studies reported blinding either the participants or the investigators to the conditions. It

should be acknowledged that a blinding of the participants is often not possible due to the nature of the interventions.

The sample sizes ranged from 12 to 217 participants ($M = 52.18$, $SD = 51.76$), and seven studies were conducted with less than 20 participants. Although there are no fixed criteria for sample size requirements, studies with participants in the range of 20 or below are considered adequate to identify usability issues for usability research purposes (Caine, 2016), but these numbers are small compared to behaviour change interventions from other domains (Norman et al., 2007).

The most frequent duration that an intervention was active for was 7 days, with a mean duration of 14.6 days ($SD = 9.60$). For six

TABLE 6 Study duration in days for the baseline measurement (Pre), the active period (At) and the post-intervention period (Post)

ID	Duration			Design	n	Recruitment criteria
	Pre	At	Post			
#1	7	7	-	WI	14	Undergrad students; high or low multitaskers.
#2a	7	14	-	WI	35	Interest in 'improving' smartphone use.
#2b				WI	27	Interest in 'improving' smartphone use.
#3a	7	35	-	WI	19	Willingness to decrease smartphone use.
#3b						
#4	7	7	-	WI	23	Android smartphone users
#5a	10	20	10	WI	12	Interest in self-tracking and productivity enhancement.
#5b					12	
#6	-	7	-	BTW & WI	104	University students
#7a	2	2	-	WI	61	Office workers and students
#7b	2	2	-	BTW & WI	55	Students from a computer mediated communication class
#8	7	14	-	BTW	36	University students who 'were willing to improve their productivity'
#9a	32	32	-	WI	217	Online users
#9b	10	10	-	WI	93	Online users
#10a	7	7	7	WI	19	Crowdworkers
#10b				WI	15	Crowdworkers
#11a	7	7	-	WI	36	Students who wanted to reduce smartphone use
#11b						
#11c						
#12a	7	21	-	WI	44	Students who wanted to reduce smartphone use
#12b						
#12c						
#13a #13b	5	10	-	WI	15	Office workers
#14	-	10	-	BTW & WI	161	Undergrad students
#15	21	30	-	BTW	73	Smartphone users
#16a	14	14	14	BTW	39	Students often distracted by Facebook
#16b					38	

Note: Whether the study design was a between subjects (BTW) design or a within subjects (WI) design. The table also contains the sample size (n), and the criteria that were used for participant recruitment. Cells with one row for multiple interventions indicate that both interventions were used in the same study.

interventions² we found reports about the post-intervention period.

For 10 interventions, the reporting of results did not allow a conversion into effect sizes.³ In seven of these cases, this conversion was not possible due to data that was not reported.⁴ For the interventions #11a, #11b, and #11c, the measure was prevented app openings, and, since the baseline period had zero prevented openings, no comparison with a baseline was possible.

One publication made their data publicly available via the servers of the Open Science Foundation (Lyngs et al., 2020).

6 | DISCUSSION

In this review, we summarized and categorized the existing evidence for digital self-control interventions. We categorized the interventions

according to their features into the categories awareness, goal-advancement, blocking, feature modification, and reward and punishment. Awareness features, which were present in 16 interventions, were the most frequent. Goal setting features were present in 10 interventions, mostly represented by time-based goal setting. Nine interventions had features to block access to distracting content. Five interventions modified the content of websites, and two variants of a single intervention used a reward scheme.

To address the research question regarding the effectiveness of the interventions, we first categorized the outcomes into the categories of time spent on distractions, the start of distractions, time spent on the device, and use of the intervention itself. While positive outcomes were reported for all types of interventions, the mixture of features within some of the interventions often made it difficult to pinpoint effects of individual features. This is especially true for features from the awareness category, such as usage statistics, which are

often present alongside other features. We assume that the awareness features are so clearly over-represented because it is comparatively easy to implement them. Information about time distribution on the device has to be collected for practically every intervention, one can create pretty visualizations out of these statistics, and it is thus tempting to include them. Presumably, these features do no harm, but they make it difficult to establish the degree to which each feature contributed to the effectiveness of an intervention. Nevertheless, several observations from the results stand out and should be considered for future interventions.

6.1 | Awareness and self-monitoring are insufficient

The results from those interventions which only contained awareness features indicate that relying solely on the users monitoring and adapting their own behaviour is not effective. This reflects the insight that people often avoid monitoring themselves, especially regarding uncomfortable topics (Chang et al., 2017). Moreover, several studies also reported greater effects for the beginning of their intervention, which suggests that a novelty effect could be at least partially responsible for the effectiveness of interventions (Tsai et al., 2019), while habitual behaviour may gain the upper hand again in the future. Whether it is through deliberate ignoring or habituation, it appears that interventions that users can easily ignore do not achieve the desired results.

6.2 | Circumvention difficulty can tip the scales

On the other hand, interventions that are more insistent in grabbing the users' attention at times of excessive distraction use are more effective. The insistence with which an intervention attempts to convince the users, and the ease with which participants can dismiss interventions appear to be relevant factors for the success of interventions. Against the background of autonomy needs and reactance (Ryan & Deci, 2000), it makes sense to give users a way to influence the strictness of interventions, and the results of those studies where users have a choice confirm a preference for interventions that allow negotiation. At the same time, too little resistance to bypassing sanctions once again leads to ignoring the interventions completely. This balancing act requires the ability to have adaptive sanctioning strictness (Schwartz et al., 2021).

6.3 | Detection of context

Although the majority of the interventions were tested on students, none of the included studies was conducted explicitly in a learning context. To confirm the applicability in learning situations, future interventions should take the context into consideration. A simple way to do this would be to ask the users what they are currently doing. A less

interruptive approach could be to monitor learning activity in digital learning environments, and infer active learning from there.

However, even when the learning context is known, distracting content cannot always be identified with certainty, when we consider learners that may need to communicate with their peers on a social media site or watch learning-related content on the same platform where they watch entertaining videos. Thus, in addition to reliable context detection, the classification into distracting or learning relevant content needs to become more sophisticated, beyond simple URL- or app-based blacklists.

6.4 | Interventions for specific groups

All the interventions were used with student or worker populations, and none with the especially-susceptible group of adolescents (van der Schuur et al., 2015). Considering that this is an especially at-risk population, they should be the target of future research (E. J. Lee & Ogbolu, 2018).

Similarly, there are differences in traits between people that may predispose them to respond better or worse to specific features (Mark et al., 2018). Those factors have to be identified for adaptive interventions.

6.5 | Long-term effects

Another topic that needs to be addressed is how well the results from an intervention transfer to longer lasting behaviour changes. Those interventions in this review that address this question with follow-ups showed no longer lasting changes.

6.6 | Limitations

One limitation that is inherent to reviews is that publications are missed due to the inclusion criteria. Many of these interventions, such as group-based locking of smartphones (Kim, Jung, et al., 2017), using smartwatches (Dibia, 2016) or a physical doll that reminds of excessive usage (Choi et al., 2016), seemed promising or used innovative approaches. We also did not incorporate interviews and other forms of qualitative analysis in this review, which often provide more nuances regarding the intervention and its reception. For those interested in a particular kind of intervention, these additional insights should be considered.

7 | CONCLUSION

There is no reason to assume that digital content will become less distracting, and that the attempts to grab our attention at every waking moment will lessen. Digital self-control interventions are one of the ways to address this problem, but there remain a lot of open issues before they can be called a solution.



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CONFLICT OF INTEREST

The authors declare that there is no conflict of interest.

ENDNOTES

¹ <https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools>.

² #5a, #5b, #16a, #16b, #10a, #10b.

³ #3a, #3b, #5a, #5b, #9a, #9b, #7b, #11a, #11b, #11c.

⁴ #3a, #3b, #5a, #5b, #7b, #9a, #9b.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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REFERENCES

- Baumgartner, S. E., Lemmens, J. S., Weeda, W. D., & Huizinga, M. (2017). Measuring media multitasking: Development of a short measure of media multitasking for adolescents. *Journal of Media Psychology*, 29(2), 1–10. <https://doi.org/10.1027/1864-1105/a000167>
- Bayer, J. B., & LaRose, R. (2018). Technology habits: Progress, problems, and prospects. In B. Verplanken (Ed.), *The psychology of habit*. Springer International Publishing. https://doi.org/10.1007/978-3-319-97529-0_7
- Caine, K. (2016, May 7). Local standards for sample size at CHI. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. San Jose, CA: ACM. <https://doi.org/10.1145/2858036.2858498>
- Chang, B. P. I., Webb, T. L., & Benn, Y. (2017). Why do people act like the proverbial ostrich? Investigating the reasons that people provide for not monitoring their goal progress. *Frontiers in Psychology*, 8. <https://doi.org/10.3389/fpsyg.2017.00152>
- Choi, S., Jeong, H., Ko, M., & Lee, U. (2016). LockDoll: Providing ambient feedback of smartphone usage within social interaction. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems - CHI EA'16*. San Jose, CA: ACM Press. <https://doi.org/10.1145/2851581.2892445>
- Coyne, M. A., & Wright, J. P. (2014). The stability of self-control across childhood. *Personality and Individual Differences*, 69, 144–149. <https://doi.org/10.1016/j.paid.2014.05.026>
- Dibia, V. (2016, October 23). FOQUS: A smartwatch application for individuals with ADHD and mental health challenges. In *Proceedings of the 18th International ACM SIGACCESS Conference on Computers and Accessibility*. Reno, NV, Association for Computing Machinery. <https://doi.org/10.1145/2982142.2982207>
- Duckworth, A. L., Taxer, J. L., Eskreis-Winkler, L., Galla, B. M., & Gross, J. J. (2019). Self-control and academic achievement. *Annual Review of Psychology*, 70(1), 373–399. <https://doi.org/10.1146/annurev-psych-010418-103230>
- Eyal, N. (2014). *Hooked: How to build habit-forming products*. New York, Portfolio/Penguin.
- Fiorella, L. (2020). The science of habit and its implications for student learning and well-being. *Educational Psychology Review*, 32, 603–625. <https://doi.org/10.1007/s10648-020-09525-1>
- Foulonneau, A., Calvary, G., & Villain, E. (2016). Stop procrastinating: TILT, time is life time, a persuasive application. In *Proceedings of the 28th Australian Conference on Computer-Human Interaction - OzCHI'16*. Launceston, Australia, ACM Press. <https://doi.org/10.1145/3010915.3010947>
- Hiniker, A., Hong, S., Kohno, T., & Kientz, J. A. (2016, May 7). MyTime: Designing and evaluating an intervention for smartphone non-use. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. San Jose, CA: ACM. <https://doi.org/10.1145/2858036.2858403>
- Holte, A. J., & Ferraro, F. R. (2020). True colors: Grayscale setting reduces screen time in college students. *The Social Science Journal*, 1–17. <https://doi.org/10.1080/03623319.2020.1737461>
- Kim, I., Jung, G., Jung, H., Ko, M., & Lee, U. (2017). Let's FOCUS: Mitigating mobile phone use in college classrooms. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 1(3), 1–29. <https://doi.org/10.1145/3130928>
- Kim, J., Cho, C., & Lee, U. (2017). Technology supported behavior restriction for mitigating self-interruptions in multi-device environments. In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (Vol. 1(3), pp. 64:1–64:21). Association for Computing Machinery. <https://doi.org/10.1145/3130932>
- Kim, J., Jung, H., Ko, M., & Lee, U. (2019). GoalKeeper: Exploring interaction lockout mechanisms for regulating smartphone use. In *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* (Vol. 3(1), pp. 16:1–16:29). Association for Computing Machinery. <https://doi.org/10.1145/3314403>
- Kim, J., Park, J., Lee, H., Ko, M., & Lee, U. (2019). LocknType: Lockout task intervention for discouraging smartphone app use. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems - CHI'19*. Glasgow, UK: ACM Press. <https://doi.org/10.1145/3290605.3300927>
- Kim, Y.-H., Jeon, J. H., Choe, E. K., Lee, B., Kim, K., & Seo, J. (2016, May 7). TimeAware: Leveraging framing effects to enhance personal productivity. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. San Jose, CA: ACM. <https://doi.org/10.1145/2858036.2858428>
- Kirschner, P. A., & Karpinski, A. C. (2010). Facebook R and academic performance. *Computers in Human Behavior*, 26(6), 1237–1245. <https://doi.org/10.1016/j.chb.2010.03.024>
- Ko, M., Yang, S., Lee, J., Heizmann, C., Jeong, J., Lee, U., Shin, D., Yatani, K., Song, J., & Chung, K.-M. (2015, February 28). NUGU: A group-based intervention app for improving self-regulation of limiting smartphone use. In *Proceedings of the 18th ACM conference on Computer Supported Cooperative Work & Social Computing*. Vancouver, Canada: Association for Computing Machinery. <https://doi.org/10.1145/2675133.2675244>
- Kovacs, G., Wu, Z., & Bernstein, M. S. (2018). Rotating online behavior change interventions increases effectiveness but also increases attrition. *Proceedings of the ACM on Human-Computer Interaction*, 2, 1–25. <https://doi.org/10.1145/3274364>
- Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. *Frontiers in Psychology*, 4. <https://doi.org/10.3389/fpsyg.2013.00863>
- Lee, E. J., & Ogbolu, Y. (2018). Does parental control work with smartphone addiction?: A cross-sectional study of children in South Korea. *Journal of Addictions Nursing*, 29(2), 128–138. <https://doi.org/10.1097/JAN.0000000000000222>
- Lee, H., Ahn, H., Choi, S., & Choi, W. (2014). The SAMS: Smartphone addiction management system and verification. *Journal of Medical Systems*, 38(1). <https://doi.org/10.1007/s10916-013-0001-1>

- Loid, K., Täht, K., & Rozgonjuk, D. (2020). Do pop-up notifications regarding smartphone use decrease screen time, phone checking behavior, and self-reported problematic smartphone use? Evidence from a two-month experimental study. *Computers in Human Behavior*, 102, 22–30. <https://doi.org/10.1016/j.chb.2019.08.007>
- Lottridge, D., Marschner, E., Wang, E., Romanovsky, M., & Nass, C. (2012). Browser design impacts multitasking. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 56(1), 1957–1961. <https://doi.org/10.1177/1071181312561289>
- Lyngs, U., Lukoff, K., Slovak, P., Binns, R., Slack, A., Inzlicht, M., Van Kleek, M., & Shadbolt, N. (2019, May 2). Self-control in cyberspace: Applying dual systems theory to a review of digital self-control tools. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. Glasgow, UK: Association for Computing Machinery. <https://doi.org/10.1145/3290605.3300361>
- Lyngs, U., Lukoff, K., Slovak, P., Seymour, W., Webb, H., Jirotko, M., Van Kleek, M., & Shadbolt, N. (2020). 'I just want to hack myself to not get distracted': Evaluating design interventions for self-control on facebook. *arXiv:2001.04180 [cs]*, arxiv 2001.04180. <https://doi.org/10.1145/3313831.3376672>
- Mark, G., Czerwinski, M., & Iqbal, S. T. (2018). Effects of individual differences in blocking workplace distractions. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems - CHI'18*. Montreal, Canada: ACM Press. <https://doi.org/10.1145/3173574.3173666>
- Masood, A., Luqman, A., Feng, Y., & Ali, A. (2020). Adverse consequences of excessive social networking site use on academic performance: Explaining underlying mechanism from stress perspective. *Computers in Human Behavior*, 113, 106476. <https://doi.org/10.1016/j.chb.2020.106476>
- Mazar, A., & Wood, W. (2018). Defining habit in psychology. In B. Verplanken (Ed.), *The psychology of habit*. Springer International Publishing. https://doi.org/10.1007/978-3-319-97529-0_2
- Moher, D., Liberati, A., Tetzlaff, J., Altman, D. G., & The PRISMA Group. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *PLoS Medicine*, 6(7), e1000097. <https://doi.org/10.1371/journal.pmed.1000097>
- Norman, G. J., Zabinski, M. F., Adams, M. A., Rosenberg, D. E., Yaroch, A. L., & Atienza, A. A. (2007). A review of eHealth interventions for physical activity and dietary behavior change. *American Journal of Preventive Medicine*, 33(4), 336–345.e16. <https://doi.org/10.1016/j.amepre.2007.05.007>
- Okeke, F., Sobolev, M., Dell, N., & Estrin, D. (2018). Good vibrations: Can a digital nudge reduce digital overload? In *Proceedings of the 20th International Conference on Human-Computer Interaction with Mobile Devices and Services - MobileHCI'18*. Barcelona, Spain: ACM Press. <https://doi.org/10.1145/3229434.3229463>
- Ophir, E., Nass, C., & Wagner, A. D. (2009). Cognitive control in media multitaskers. *Proceedings of the National Academy of Sciences of the United States of America*, 106(37), 15583–15587. <https://doi.org/10.1073/pnas.0903620106>
- Parry, D. A., & le Roux, D. B. (2019). Media multitasking and cognitive control: A systematic review of interventions. *Computers in Human Behavior*, 92, 316–327. <https://doi.org/10.1016/j.chb.2018.11.031>
- Petticrew, M., & Roberts, H. (Eds.). (2006). *Systematic reviews in the social sciences*. Blackwell Publishing Ltd. <https://doi.org/10.1002/9780470754887>
- Prochaska, J. O., & Velicer, W. F. (1997). The transtheoretical model of health behavior change. *American Journal of Health Promotion*, 12(1), 38–48. <https://doi.org/10.4278/0890-1171-12.1.38>
- Rosen, L. D., Mark Carrier, L., & Cheever, N. A. (2013). Facebook and texting made me do it: Media-induced task-switching while studying. *Computers in Human Behavior*, 29(3), 948–958. <https://doi.org/10.1016/j.chb.2012.12.001>
- Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68–78. <https://doi.org/10.1037/0003-066X.55.1.68>
- Schwartz, R., Monge Roffarello, A., De Russis, L., & Apostolellis, P. (2021, May 8). Reducing risk in digital self-control tools: Design patterns and prototype. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems. CHI'21*. Yokohama, Japan: ACM. <https://doi.org/10.1145/3411763.3451843>
- Shea, N., Boldt, A., Bang, D., Yeung, N., Heyes, C., & Frith, C. D. (2014). Supra-personal cognitive control and metacognition. *Trends in Cognitive Sciences*, 18(4), 186–193. <https://doi.org/10.1016/j.tics.2014.01.006>
- Shenhav, A., Botvinick, M. M., & Cohen, J. D. (2013). The expected value of control: An integrative theory of anterior cingulate cortex function. *Neuron*, 79(2), 217–240. <https://doi.org/10.1016/j.neuron.2013.07.007>
- Terry, C. A., Mishra, P., & Roseth, C. J. (2016). Preference for multitasking, technological dependency, student metacognition, & pervasive technology use: An experimental intervention. *Computers in Human Behavior*, 65, 241–251. <https://doi.org/10.1016/j.chb.2016.08.009>
- Tsay, C. H.-H., Kofinas, A. K., Trivedi, S. K., & Yang, Y. (2019). Overcoming the novelty effect in online gamified learning systems: An empirical evaluation of student engagement and performance. *Journal of Computer Assisted Learning*, 36, 128–146. <https://doi.org/10.1111/jcal.12385>
- Tseng, V. W.-S., Lee, M. L., Denoue, L., & Avrahami, D. (2019). Overcoming distractions during transitions from break to work using a conversational website-blocking system. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems - CHI'19*. Glasgow, UK: ACM Press. <https://doi.org/10.1145/3290605.3300697>
- van der Schuur, W. A., Baumgartner, S. E., Sumter, S. R., & Valkenburg, P. M. (2015). The consequences of media multitasking for youth: A review. *Computers in Human Behavior*, 53, 204–215. <https://doi.org/10.1016/j.chb.2015.06.035>
- Varol, F., & Yildirim, E. (2019). Cyberloafing in higher education: Reasons and suggestions from students' perspectives. *Technology, Knowledge and Learning*, 24(1), 129–142. <https://doi.org/10.1007/s10758-017-9340-1>
- Whittaker, S., Kalnikaite, V., Hollis, V., & Guydish, A. (2016, May 7). 'Don't waste my time': Use of time information improves focus. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. San Jose, CA: Association for Computing Machinery. <https://doi.org/10.1145/2858036.2858193>
- Willems, Y. E., Boesen, N., Li, J., Finkenauer, C., & Bartels, M. (2019). The heritability of self-control: A meta-analysis. *Neuroscience and Biobehavioral Reviews*, 100, 324–334. <https://doi.org/10.1016/j.neubiorev.2019.02.012>
- Wood, W., & Rünger, D. (2016). Psychology of habit. *Annual Review of Psychology*, 67(1), 289–314. <https://doi.org/10.1146/annurev-psych-122414-033417>

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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